

Data Scientist | Data Analyst | Machine Learning Engineer

Personal Information

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PROFESSIONAL SUMMARY

Data Scientist with a background in Applied Mathematics and a Master's degree in Astrophysics, with one year of technical experience in Data Science and two years in Data Analytics. I have developed data science solutions for classification, regression, time series forecasting, natural language processing, and computer vision using Python, SQL, Excel, pandas, NumPy, SciPy, scikit-learn, TensorFlow, CatBoost, Git, and Bash. In data analytics, I have worked on data acquisition, validation, quality control, processing, and large-scale data analysis, automating pipelines that reduced processing time by up to 95%. I also evaluate the performance of artificial intelligence models based on reasoning, accuracy, instruction-following, and quality standards. My background in Applied Mathematics and Astrophysics strengthens my ability to analyze problems, build models, and interpret results critically, delivering data-driven solutions that create measurable business impact.

TECHNICAL SKILLS

Programming Languages: Python | SQL | Bash | Java | JavaScript | C#

Python Libraries: pandas | NumPy | SciPy | Statsmodels | scikit-learn | TensorFlow | CatBoost | LightGBM | XGBoost | Matplotlib | Seaborn | Plotly | OpenCV | NLTK | spaCy | Hugging Face Transformers | Requests | BeautifulSoup | Selenium | openpyxl | XlsxWriter | pytest | Joblib

Machine Learning & AI: Classification | Regression | Time Series Forecasting | Natural Language Processing | Computer Vision | Deep Learning | LLM Evaluation | Prompt Engineering | Function Calling

Evaluation Metrics: ROC-AUC | F1 Score | Precision | Recall | Accuracy | RMSE | MAE | R^2 Score | Confusion Matrix

Statistics: Descriptive Statistics | Inferential Statistics | Hypothesis Testing | A/B Testing | Bootstrap | Sampling | Probability Distributions | Confidence Intervals | Correlation | Linear Regression | Logistic Regression

Mathematics: Linear Algebra | Probability | Statistics | Stochastic Processes | Optimization | Differential Calculus | Integral Calculus | Multivariable Calculus | Ordinary Differential Equations | Partial Differential Equations | Numerical Analysis | Complex Analysis | Real Analysis | Measure Theory

Databases: PostgreSQL | MySQL

Data Visualization & BI: Power BI | Excel | Matplotlib | Seaborn | Plotly

Data Engineering & Automation: ETL | Web Scraping | REST APIs

Tools: Git | GitHub | Jupyter Notebook | Google Colab | Visual Studio Code | LaTeX | Overleaf | Microsoft Office

Operating Systems: Linux | Windows

TECHNICAL EXPERIENCE

Data Scientist | TripleTen

May 2025 – May 2026

- Developed more than 6 end-to-end data science projects, from data collection, validation, and analysis to the development and evaluation of machine learning models.
- Built solutions for classification, regression, time series forecasting, natural language processing, and computer vision tasks.
- Processed and analyzed datasets containing up to 4.5 million records, applying statistical analysis and data visualization to identify patterns, trends, and business opportunities.
- Developed a customer churn prediction model that achieved a ROC-AUC of 0.925, exceeding the project's target of 0.81 by 14.2%.
- Managed version control and technical documentation using Git and GitHub.

PROFESSIONAL EXPERIENCE

Artificial Intelligence Analyst | Outlier

Dec 2024 – Present

- Evaluate responses generated by language models, assessing reasoning, accuracy, instruction compliance, and safety.
- Review and validate the work of other evaluators as a Trusted Reviewer and Oracle, ensuring compliance with each project's quality standards.
- Write and evaluate mathematics, programming, and logical reasoning problems to measure the performance of artificial intelligence models.
- Assess the ability of language models to select and invoke functions through APIs and external tools.
- Provide technical feedback to identify recurring errors and support continuous model improvement.

Data Analyst | MEGASTAR Project, INAOE

May 2023 – Dec 2024

- Automated the large-scale execution of the data reduction pipeline using Python and Bash, reducing processing time by 95%.
- Processed, validated, and cleaned more than 2,000 high-resolution spectroscopic observations, exceeding the project's initial target by more than four times for inclusion in a public data library.
- Designed and implemented quality control procedures to detect inconsistencies, validate metadata, and ensure data integrity prior to publication.
- Integrated automation scripts into the project's workflow to standardize pipeline execution for other collaborators.

University Mathematics Tutor | Multiple Institutions

2016 – 2021

- Provided university-level tutoring in probability, statistics, calculus, and linear algebra to students from Tecnológico de Monterrey, La Salle, IPN, and UAEH, adapting instruction to different levels and learning objectives.
- Designed and evaluated customized exercises focused on quantitative analysis, mathematical modeling, and data interpretation, strengthening students' understanding of statistical concepts.

PROJECTS

Customer churn prediction for Interconnect | 2026

github.com/marcofrancohalsey/interconnect-churn-prediction

- Description: Developed a machine learning model to predict customer churn for a telecommunications company, with the goal of identifying users at higher risk of leaving.
- Stack: Python, pandas, NumPy, scikit-learn, CatBoost, Matplotlib, Seaborn, SciPy, Jupyter Notebook, Git.
- Impact: The model achieved a ROC-AUC score of 0.9253 on the test set, exceeding the project target of 0.81 and demonstrating strong performance in identifying customers at risk of churn.

Taxi demand forecasting using time series | 2025

github.com/marcofrancohalsey/hourly-taxi-demand-forecasting

- Description: Developed a machine learning model to forecast taxi demand one hour ahead using historical data, enabling demand anticipation and supporting driver allocation.
- Stack: Python, pandas, NumPy, scikit-learn, Linear Regression, Random Forest, LightGBM, CatBoost, Statsmodels, Matplotlib, Jupyter Notebook, Git.
- Impact: The model achieved an RMSE of 34.94 requests per hour on the test set, outperforming the project target of 48 and reducing prediction error by 8.92 to 12.47 requests per hour compared with Random Forest, LightGBM, and CatBoost.

EDUCATION

Bachelor of Science in Applied Mathematics Autonomous University of the State of Hidalgo	2015 – 2021
Master of Science in Astrophysics National Institute of Astrophysics, Optics and Electronics (INAOE)	2022 – 2024
Data Science Bootcamp TripleTen	2025 – 2026

CERTIFICATIONS

- Web Development with HTML & CSS | CodeHS | 2021
- SQL Programming | CodeHS | 2021
- Computer Science in Python 3 | CodeHS | 2020
- Computer Science in JavaScript (Golden) | CodeHS | 2020
- Computer Science in Java | CodeHS | 2019

LANGUAGES

Spanish (Native) | English B2 – Upper Intermediate